Professional Elective – I

ARTIFICIAL INTELLIGENCE

(Common to CSE & IT)
III Year – I Semester

Lecture : 3	Internal Marks : 30
Credits : 3	External Marks : 70

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Course Objectives

- To learn the methods of solving problems using Artificial Intelligence.
- To introduce the concepts of Expert Systems.

Course Outcomes

Upon successful completion of the course, the students will be able to

- formulate problems using Artificial Intelligence representations.
- evaluate the effectiveness of heuristic search and game playing algorithms in different domains.
- design and implement knowledge bases that combine logical, semantic, and probabilistic representations.
- apply resolution and explanation-based learning techniques to derive conclusions from logical premises.
- analyze case studies of prominent expert systems like MYCIN, DART, and XCON.

Course Content

UNIT – I:

Introduction: AI problems, foundation of AI and history of AI intelligent agents: Agents and Environments, the concept of rationality, the nature of environments, structure of agents, problem solving agents, problem formulation.

UNIT – II:

Searching- Searching for solutions, uniformed search strategies – Breadth first search, depth first Search. Search with partial information (Heuristic search) Hill climbing, A*, AO* Algorithms, Problem reduction, Game Playing-Adversial search, Games, mini-max algorithm, optimal decisions in multiplayer games, Problem in Game playing, Alpha-Beta pruning, Evaluation functions.

UNIT – III:

Representation of Knowledge: Knowledge representation issues, predicate logic- logic programming, semantic nets- frames and inheritance, representing knowledge using rules. Reasoning under uncertainty, review of probability, Bayes' probabilistic interferences and dempster shafer theory.

UNIT – IV:

Logic concepts: First order logic. Inference in first order logic, propositional vs. first order inference, unification & lifts forward chaining, Backward chaining, Resolution, Learning from observation Inductive learning, Decision trees, Explanation based learning, Reinforcement Learning.

UNIT – V:

Expert Systems: Architecture of expert systems, Roles of expert systems – Knowledge Acquisition Meta knowledge Heuristics. Typical expert systems – MYCIN, DART, XCON: Expert systems shells.